MILITARY SPECIFICATION SHEET

CABLE, ELECTRICAL, TYPE LS1SU
(INCLUDING VARIATION LS1SUA)

This specification sheet is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for acquiring the cable described herein shall consist of this document and the latest issue of MIL-C-24643.

This cable can be manufactured in two variations: LS1SU (unarmored) and LS1SUA (armored).

REQUIREMENTS:

Qualification required.

Construction (non-watertight)

First - Copper conductor, tin coated, in accordance with ASTM B 286, size 20-7 (maximum diameter 0.039 inch), and AWG no. 18, class B stranding (7 by 0.0152 inch) (maximum diameter 0.050 inch). For LS1SU-36, 32 size 20-7 and 4 AWG no. 18 conductors. For LS1SU-60, 60 size 20-7 conductors.

Second - Separator may be used at manufacturer's option where required to provide free-stripping insulation.

Third - Cross-linked polyethylene insulation. Minimum wall thickness 17 mils.

Fourth - Braided shield of AWG no. 36 uncoated copper. Braid angle of 30 to 35 degrees; minimum coverage of 85 percent.

Fifth - Shield insulation of two polyester tapes, type G in accordance with MIL-I-631, sealed. Standard identification code applied by method 2, on inner tape. The four size AWG no. 18 conductors shall be numbers 1 through 4.

Sixth - The required number of singles cabled with a lay not greater than 15 times the pitch diameter of the layer. Cabling sequence shall be consecutive, starting with no. 1, from center outward.

Seventh - Binder tape applied helically with overlap.

FSC 6145

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Construction variant for type LS1SU

Eighth - Cross-linked polyolefin jacket. (See table I for thickness). Cable surface marking required.

Construction variant for type LS1SUA

Eighth - Cross-linked polyolefin jacket. (See table I for thickness).
Ninth - Braided metal armor and paint.

TABLE I. Details.

Military 1/ part no. M24643/42	Size 2/	Conductor size AWG no.	Diameter over completed conductors nominal (inch)	Cable jacket thickness minimum (inch)	Overall diameter (inches)			
					Type LS1SU		Type LS1SUA	
					min	max	min	max
-01UN	36	18 20-7	0.127 .115	0.060	0.910	0.985	0.960	1.035
-02UN	60	20-7	.115	.080	1.210	1.310	1.260	1.360

1/ Unarmored. When armor option is required, see MIL-C-24643 for configuration of part number.
1/ Type LS1SU and LS1SUA.

EXAMINATION AND TESTS:

	Requirements
Basic electrical:	
Conductor resistance - ohms/1000 feet, at 25°C, maximum.	
ASTM B 286, size 20-7	10.93
Size AWG no. 18 (class B)	7.05
Voltage withstand - volts, root mean square, minimum	
Conductor to shield	2200
Shield to shield	500
Insulation resistance - megohms/1000 feet, minimum	500
Conductor continuity	No failure
Group A:	
Visual and dimensional	No failure
Capacitance - at 1 kHz, pF/ft, maximum	72

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EXAMINATION AND TESTS: (Continued)

	Requirements
Group B:	
Cold bending, cable - at minus 30 ± 2°C, maximum mandrel diameter, 12 times maximum cable diameter	No damage
Drip - 95 ± 1°C	Zero
Tear - pounds per inch thickness, minimum (ASTM D 470)	35
Physicals (unaged)	
Insulation	
Tensile strength - lb/in^2, minimum	1800
Elongation - percent, minimum	250
Jacket (cable)	
Tensile strength - lb/in^2, minimum	1300
Elongation - percent, minimum	160
Group C:	
Physicals (aged)	
Insulation	
Air oven	
Tensile strength - percent of unaged, minimum	80
Elongation - percent of unaged, minimum	80
Jacket (cable)	
Air oven	
Tensile strength - percent of unaged, minimum	60
Elongation - percent of unaged, minimum	60
Hot oil immersion	
Tensile strength - lb/in^2, minimum	50
Elongation - percent, minimum	50
Shrinkage	No failure
Permanence of printing (jacket) type LS1SU only - cycles, minimum	250
Heat distortion - percent of unaged, maximum	30
Cable filler removability	No failure
Armor (type LS1SUA only) - conformance to material, construction and coverage	No failure
Shield - conformance to material, construction and coverage	No failure

QUALIFICATION INSPECTION:

Qualification inspection shall include basic electrical, all of groups A, B and C, plus the following:

Aging and compatibility (cable) (125 ± 5°C)	No failure
Abrasion resistance (jacket/insulation)	No failure
Acid gas equivalent - percent, maximum	
Jacket	2
Fillers	2
Insulation	18

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QUALIFICATION INSPECTION: (Continued)

	Requirements
Halogen content - percent, maximum	
Jacket	0.2
Fillers	0.2
Flame propagation (cable)	No failure
Immersion (jacket)	
Tensile strength - percent of unaged, minimum	50
Elongation - percent of unaged, minimum	50
Smoke index, maximum	
Jacket	25
Fillers	—
Insulation	10
Toxicity index, maximum	
Jacket	5
Fillers	5
Insulation	1.5
Durometer (jacket) - (type A) hardness, minimum	80
Weathering (jacket)	No failure

UNIT ORDERING LENGTHS:
All sizes 1000 feet (nominal)

NOTE: Not for Air Force use.

Custodians:
Army - MI
Navy - SH

Preparing activity:
Navy - SH
(Project 6145-0959)

Review activities:
Army - AV, CR, ER
Navy - EC
DLA - IS

User activities:
Army - ME, AR, AL
Navy - CG

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